Tutorial: fibrational perspectives on logic and language

Part II: bifibrations

Noam Zeilberger

Ecole Polytechnique (Palaiseau, France)

RAMiCS 2026 @ Poznán, 8 April 2026

3. Bifibrations in logic and language

What is a bifibration?

Formally:

$$\begin{tikzcd} \mathcal{E} \arrow[d, "\pi"'] \\ \mathcal{B} \end{tikzcd}$$

What is a bifibration?

Formally:

$$\begin{tikzcd} \mathcal{E} \arrow[d, "\pi"'] & S \arrow[d, dotted, no head] & \\ \mathcal{B} & A \arrow[r, "f"] & B \end{tikzcd}$$

What is a bifibration?

Formally:

$$\begin{tikzcd} \mathcal{E} \arrow[d, "\pi"'] & S \arrow[r, dashed, "f_S"] \arrow[d, dotted, no head] & f^{+}S \arrow[d, dotted, no head] \\ \mathcal{B} & A \arrow[r, "f"] & B \end{tikzcd}$$

What is a bifibration?

Formally:

$$\begin{tikzcd} \mathcal{E} \arrow[d, "\pi"'] \\ \mathcal{B} \end{tikzcd} \qquad \begin{tikzcd} S \arrow[r, dashed, "f_S"] \arrow[d, dotted, no head] & f^{+}S \arrow[d, dotted, no head] \\ A \arrow[r, "f"] & B \end{tikzcd} \qquad \begin{tikzcd} & T \arrow[d, dotted, no head] \\ B \arrow[r, "g"] & C \end{tikzcd}$$

What is a bifibration?

Formally:

$$\begin{tikzcd} \mathcal{E} \arrow[d, "\pi"'] \\ \mathcal{B} \end{tikzcd} \qquad \begin{tikzcd} S \arrow[r, dashed, "f_S"] \arrow[d, dotted, no head] & f^{+}S \arrow[d, dotted, no head] \\ A \arrow[r, "f"] & B \end{tikzcd} \qquad \begin{tikzcd} g^{-}T \arrow[r, dashed, "\overline{g}_T"] \arrow[d, dotted, no head] & T \arrow[d, dotted, no head] \\ B \arrow[r, "g"] & C \end{tikzcd}$$

What is a bifibration?

Formally:

$$\begin{tikzcd} \mathcal{E} \arrow[d, "\pi"'] \\ \mathcal{B} \end{tikzcd} \qquad \begin{tikzcd} S \arrow[r, dashed, "f_S"] \arrow[d, dotted, no head] & f^{+}S \arrow[d, dotted, no head] \\ A \arrow[r, "f"] & B \end{tikzcd} \qquad \begin{tikzcd} g^{-}T \arrow[r, dashed, "\overline{g}_T"] \arrow[d, dotted, no head] & T \arrow[d, dotted, no head] \\ B \arrow[r, "g"] & C \end{tikzcd}$$

...and these liftings should be universal...

What is a bifibration?

Formally:

$$\begin{tikzcd} \mathcal{E} \arrow[d, "\pi"'] \\ \mathcal{B} \end{tikzcd} \qquad \begin{tikzcd} S \arrow[r, dashed, "f_S"] \arrow[d, dotted, no head] & f^{+}S \arrow[d, dotted, no head] \\ A \arrow[r, "f"] & B \end{tikzcd} \qquad \begin{tikzcd} g^{-}T \arrow[r, dashed, "\overline{g}_T"] \arrow[d, dotted, no head] & T \arrow[d, dotted, no head] \\ B \arrow[r, "g"] & C \end{tikzcd}$$

...and these liftings should be universal...

$$\begin{tikzcd} S \arrow[rr, "\alpha"] & & T \end{tikzcd}$$

$$\begin{tikzcd} A \arrow[r, "f"] & B \arrow[r, "g"] & C \end{tikzcd}$$

What is a bifibration?

Formally:

$$\begin{tikzcd} \mathcal{E} \arrow[d, "\pi"'] \\ \mathcal{B} \end{tikzcd} \qquad \begin{tikzcd} S \arrow[r, dashed, "f_S"] \arrow[d, dotted, no head] & f^+ S \arrow[d, dotted, no head] \\ A \arrow[r, "f"] & B \end{tikzcd} \qquad \begin{tikzcd} g^- T \arrow[r, dashed, "\overline{g}_T"] \arrow[d, dotted, no head] & T \arrow[d, dotted, no head] \\ B \arrow[r, "g"] & C \end{tikzcd}$$

...and these liftings should be universal...

$$\begin{tikzcd} S \arrow[r, "f_S"] & f^+ S \arrow[r, "f \backslash_g \alpha"] & T \\ A \arrow[r, "f"] & B \arrow[r, "g"] & C \end{tikzcd} = \begin{tikzcd} S \arrow[rr, "\alpha"] & & T \\ A \arrow[r, "f"] & B \arrow[r, "g"] & C \end{tikzcd}$$

What is a bifibration?

Formally:

$$\begin{tikzcd} \mathcal{E} \arrow[d, "\pi"'] \\ \mathcal{B} \end{tikzcd} \qquad \begin{tikzcd} S \arrow[r, dashed, "f_S"] \arrow[d, dotted, no head] & f^+ S \arrow[d, dotted, no head] \\ A \arrow[r, "f"] & B \end{tikzcd} \qquad \begin{tikzcd} g^- T \arrow[r, dashed, "\overline{g}_T"] \arrow[d, dotted, no head] & T \arrow[d, dotted, no head] \\ B \arrow[r, "g"] & C \end{tikzcd}$$

...and these liftings should be universal...

$$\begin{tikzcd} S \arrow[r, "f_S"] & f^+ S \arrow[r, "f \backslash_g \alpha"] & T \\ A \arrow[r, "f"] & B \arrow[r, "g"] & C \end{tikzcd} = \begin{tikzcd} S \arrow[rr, "\alpha"] & & T \\ A \arrow[r, "f"] & B \arrow[r, "g"] & C \end{tikzcd} = \begin{tikzcd} S \arrow[r, "\alpha / f \overline{g}"] & g^- T \arrow[r, "\overline{g}_T"] & T \\ A \arrow[r, "f"] & B \arrow[r, "g"] & C \end{tikzcd}$$

Bifibrations as families of adjunctions

A consequence of the definition is that in a bifibration $\pi : \mathcal{E} \to \mathcal{B}$, pushing and pulling along any given arrow $f : A \to B$ in $\mathcal{B}$ induces an *adjunction* between fiber categories:

$$\begin{tikzcd} \mathcal{E}_A \arrow[r, bend left, "f^+"] \arrow[r, phantom, "\perp"] & \mathcal{E}_B \arrow[l, bend left, "f^-"] \end{tikzcd} \qquad \frac{S \Longrightarrow_{\mathrm{id}_A} f^- T}{\overline{\overline{f^+ S \Longrightarrow_{\mathrm{id}_B} T}}}$$

Indeed, any bifibration $\mathcal{E} \to \mathcal{B}$ can be transformed into a pseudofunctor $\mathcal{B} \to \mathrm{Adj}$, and vice versa. Thus bifibrations can be found everywhere there are adjunctions!

Example: bifibration of subsets over Set

Let Set be the category of sets and functions.

Let SubSet be the category whose objects are pairs $(A, S \subset A)$ and whose arrows $(A, S) \to (B, T)$ are functions $f : A \to B$ such that $x \in S \Longrightarrow f(x) \in T$.

The forgetful functor $\mathrm{SubSet} \to \mathrm{Set}$ is a bifibration, where

$$f^{+}(A, S) = (B,\ f(S)\ )$$
$$f^{-}(B, T) = (A,\ f^{-1}(T)\ )$$

In particular, the adjunction $f^{+} \dashv f^{-}$ reduces to the fact that

$$f(S) \subseteq T \iff S \subseteq f^{-1}(T)$$

Example: bifibration of subsets over Rel

Let Rel be the category of sets and relations.

Let SubRel be the category whose objects are pairs $(A, S \subset A)$ and whose arrows $(A, S) \to (B, T)$ are relations $r : A \nrightarrow B$ such that $x \in S \wedge (x, y) \in r \Longrightarrow y \in T$.

The forgetful functor $\mathrm{SubRel} \to \mathrm{Rel}$ is a bifibration, where

$$r^{+}(A, S) = (B,\ \{\, y \mid \exists x.(x, y) \in r \wedge x \in S \,\}\ )$$
$$r^{-}(B, T) = (A,\ \{\, x \mid \forall y.(x, y) \in r \supset y \in T \,\}\ )$$

(Compare $r^{+} \dashv r^{-}$ with Galois connection $\blacklozenge\phi \vdash \psi \iff \phi \vdash \Box\psi$ from modal logic.)

Example: Hoare logic

Let Cmd be a one-object category whose arrows $c : * \to *$ are sequential compositions of program commands defined over some global set of variables.

Let Prf be the category whose objects are predicates on the global state, and whose arrows $P \to Q$ are given by commands c that satisfy the Hoare triple $\{P\}c\{Q\}$.

The following are equivalent:

1. the evident forgetful functor $\mathrm{Prf} \to \mathrm{Cmd}$ is a bifibration;
2. *weakest preconditions* and *strongest postconditions* exist for all P, c, Q.

$$P \Rightarrow wp(c, Q) \quad \Longleftrightarrow \quad \{P\}c\{Q\} \quad \Longleftrightarrow \quad sp(c, P) \Rightarrow Q$$

Example from language theory

Recall the poset of formal languages $(P(\Sigma^*), \subseteq)$ has the structure of a residuated monoid, with concatenation and left- and right-division of languages defined by:

$$K \cdot L \stackrel{\text{def}}{=} \{\, uv \mid u \in K \wedge v \in L \,\}$$

$$K \backslash L \stackrel{\text{def}}{=} \{\, v \mid u \in K \supset uv \in L \,\}$$

$$L / K \stackrel{\text{def}}{=} \{\, u \mid v \in K \supset uv \in L \,\}$$

This is an instance of the following general construction...

Example from language theory

Proposition. Let $(A, m : A \otimes A \to A, e : 1 \to A)$ be a monoid in the base of a symmetric monoidal closed bifibration $\pi : \mathcal{E} \to \mathcal{C}$. Then the fiber $\mathcal{E}_A$ is monoidal biclosed, with tensor and left- and right-internal homs defined by

$$S \otimes_A T \stackrel{\text{def}}{=} m^+(S \otimes_{\mathcal{E}} T) \qquad S \backslash_A T \stackrel{\text{def}}{=} m_\ell^-(S \multimap T) \qquad T /_A S \stackrel{\text{def}}{=} m_r^-(S \multimap T)$$

where $m_\ell, m_r : A \to A \multimap A$ denote the left- and right-curryings of m.

Example from language theory

Proposition. Let $(A, m : A \otimes A \to A, e : 1 \to A)$ be a monoid in the base of a symmetric monoidal closed bifibration $\pi : \mathcal{E} \to \mathcal{C}$. Then the fiber $\mathcal{E}_A$ is monoidal biclosed, with tensor and left- and right-internal homs defined by

$$S \otimes_A T \stackrel{\text{def}}{=} m^+(S \otimes_{\mathcal{E}} T) \qquad S \backslash_A T \stackrel{\text{def}}{=} m_\ell^-(S \multimap T) \qquad T /_A S \stackrel{\text{def}}{=} m_r^-(S \multimap T)$$

where $m_\ell, m_r : A \to A \multimap A$ denote the left- and right-curryings of m.

Considering $(\Sigma^*, \cdot, \epsilon)$ as a monoid in the base of the cartesian closed bifibration $\mathrm{SubSet} \to \mathrm{Set}$ recovers the monoidal closed structure on $(P(\Sigma^*), \subseteq)$.

4. The logic of bifibrations

Free bifibrations

Most functors are not bifibrations, but one can try to freely turn them into bifibrations:

$$\begin{tikzcd} \mathcal{D} \arrow[d, "p"'] \\ \mathcal{C} \end{tikzcd}$$

Free bifibrations

Most functors are not bifibrations, but one can try to freely turn them into bifibrations:

$$\begin{tikzcd} \mathcal{D} \arrow[dr, "p"'] & \\ & \mathcal{C} \end{tikzcd}$$

Free bifibrations

Most functors are not bifibrations, but one can try to freely turn them into bifibrations:

$$\begin{tikzcd}
\mathcal{D} \arrow[dr, "p"'] & \\
& \mathcal{C}
\end{tikzcd}
\qquad
\begin{tikzcd}
\mathcal{B}\mathrm{if}(p) \arrow[d, "\Lambda_p"] \\
\mathcal{C}
\end{tikzcd}$$

Free bifibrations

Most functors are not bifibrations, but one can try to freely turn them into bifibrations:

$$\begin{tikzcd}
\mathcal{D} \arrow[r, "\eta_p"] \arrow[dr, "p"'] & \mathcal{B}\mathrm{if}(p) \arrow[d, "\Lambda_p"] \\
& \mathcal{C}
\end{tikzcd}$$

Free bifibrations

Most functors are not bifibrations, but one can try to freely turn them into bifibrations:

$$
\begin{tikzcd}
\mathcal{D} \arrow[r, "\eta_p"] \arrow[dr, "p"'] \arrow[drr, "\theta"] & \mathcal{B}\mathrm{if}(p) \arrow[d, "\Lambda_p"] & \\
& \mathcal{C} & \mathcal{E} \arrow[l, "q \text{ bifibration}"']
\end{tikzcd}
$$

Free bifibrations

Most functors are not bifibrations, but one can try to freely turn them into bifibrations:

$$
\begin{tikzcd}
\mathcal{D} \arrow[r, "\eta_p"] \arrow[dr, "p"'] \arrow[drr, "\theta"] & \mathcal{B}\mathrm{if}(p) \arrow[d, "\Lambda_p"] \arrow[dr, dashed, "[\![-]\!]_\theta"] & \\
& \mathcal{C} & \mathcal{E} \arrow[l, "q \text{ bifibration}"]
\end{tikzcd}
$$

Free bifibrations

Most functors are not bifibrations, but one can try to freely turn them into bifibrations:

$$
\begin{tikzcd}
\mathcal{D} \arrow[r, "\eta_p"] \arrow[dr, "p"'] \arrow[drr, "\theta"] & \mathcal{B}\mathrm{if}(p) \arrow[d, "\Lambda_p"] \arrow[dr, dashed, "[\![-]\!]_\theta"] & \\
& \mathcal{C} & \mathcal{E} \arrow[l, "q \text{ bifibration}"]
\end{tikzcd}
$$

Surprisingly, this problem has been relatively little-studied:

- R. Dawson, R. Paré, and D. Pronk (DPP). Adjoining adjoints. *Adv. Mathematics*, 2003.
- F. Lamarche. Path functors in Cat. Unpublished, 2010. HAL-00831430.

Free bifibrations

Most functors are not bifibrations, but one can try to freely turn them into bifibrations:

$$
\begin{tikzcd}
\mathcal{D} \arrow[r, "\eta_p"] \arrow[dr, "p"'] \arrow[drr, "\theta"] & \mathcal{B}\mathrm{if}(p) \arrow[d, two heads, "\Lambda_p"] \arrow[dr, dashed, "[\![-]\!]_\theta"] & \\
& \mathcal{C} & \mathcal{E} \arrow[l, "q \text{ bifibration}"]
\end{tikzcd}
$$

Surprisingly, this problem has been relatively little-studied:

- R. Dawson, R. Paré, and D. Pronk (DPP). Adjoining adjoints. *Adv. Mathematics*, 2003.
- F. Lamarche. Path functors in Cat. Unpublished, 2010. HAL-00831430.

In joint work with Bryce Clarke and Gabriel Scherer, we gave an explicit proof-theoretic construction and found some interesting examples of free bifibrations (arxiv:2511.07314).

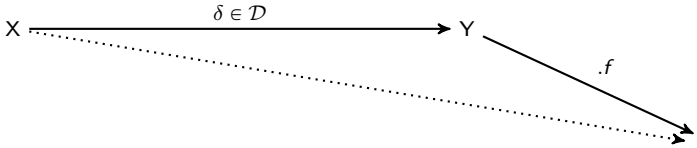
Building the free bifibration: intuition

Introduce "formal" push/pull along the arrows of $\mathcal{C}$.

$$\mathsf{X} \xrightarrow{\delta \in \mathcal{D}} \mathsf{Y}$$

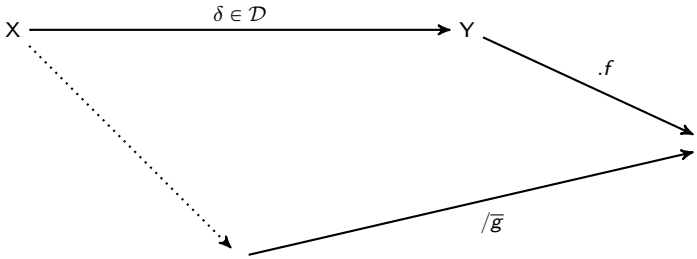
Building the free bifibration: intuition

Introduce "formal" push/pull along the arrows of $\mathcal{C}$.



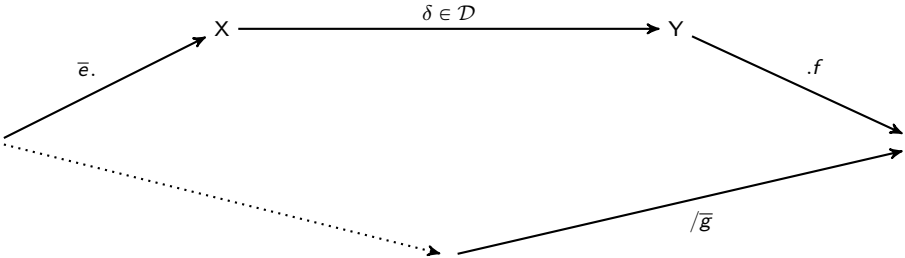
Building the free bifibration: intuition

Introduce "formal" push/pull along the arrows of $\mathcal{C}$.



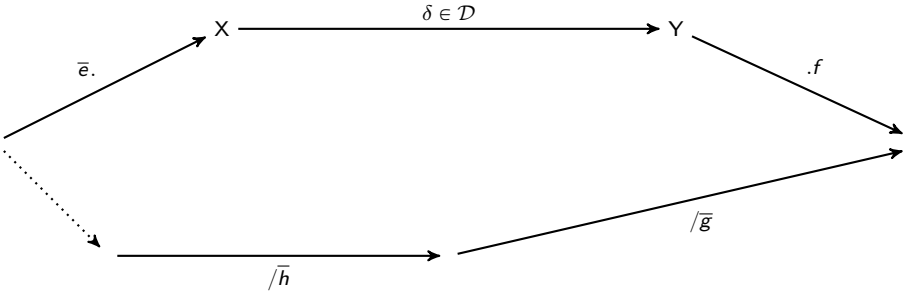
Building the free bifibration: intuition

Introduce "formal" push/pull along the arrows of $\mathcal{C}$.



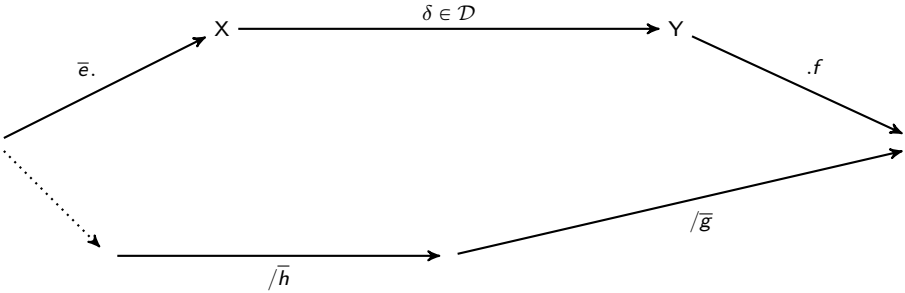
Building the free bifibration: intuition

Introduce "formal" push/pull along the arrows of $\mathcal{C}$.



Building the free bifibration: intuition

Introduce "formal" push/pull along the arrows of $\mathcal{C}$.



Some operations commute: non-trivial equivalence.

An example derivation in a free bifibration

$$\begin{tikzcd}
\mathcal{D} \arrow[d, "p"] & X \arrow[r, "\alpha"] & Y \\
\mathcal{C} & A \arrow[r, "f"] \arrow[rr, bend right, "h"'] & B \arrow[r, "g"] \arrow[d, phantom, "\equiv"] & C \\
& & {} &
\end{tikzcd}$$

An example derivation in a free bifibration

$$\begin{tikzcd}
\mathcal{D} \arrow[d, "p"] & X \arrow[r, "\alpha"] & Y \\
\mathcal{C} & A \arrow[r, "f"] \arrow[rr, bend right, "h"'] \arrow[r, phantom, shift right=3ex, "\equiv"] & B \arrow[r, "g"] & C
\end{tikzcd}$$

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{X \underset{f}{\Longrightarrow} Y}\,\alpha}{X \underset{fg}{\Longrightarrow} g^{+}\, Y}\,\mathrm{R}g^{+}}{f^{+}\, X \underset{g}{\Longrightarrow} g^{+}\, Y}\,\mathrm{L}f^{+}}{f^{-}\, f^{+}\, X \underset{fg}{\Longrightarrow} g^{+}\, Y}\,\mathrm{L}f^{-}}{\dfrac{f^{-}\, f^{+}\, X \underset{h}{\Longrightarrow} g^{+}\, Y}{f^{-}\, f^{+}\, X \underset{\mathrm{id}_A}{\Longrightarrow} h^{-}\, g^{+}\, Y}\,\mathrm{R}h^{-}}$$

An example derivation in a free bifibration

$$\begin{tikzcd}
\mathcal{D} \arrow[d, "p"] & X \arrow[r, "\alpha"] & Y \\
\mathcal{C} & A \arrow[r, "f"] \arrow[rr, bend right, "h"'] \arrow[r, phantom, shift right=3ex, "\equiv"] & B \arrow[r, "g"] & C
\end{tikzcd}$$

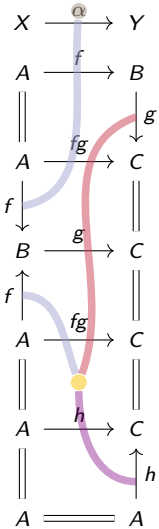
$$\dfrac{}{X \underset{f}{\Longrightarrow} Y}\ \alpha$$

$$\dfrac{}{X \underset{fg}{\Longrightarrow} g^{+}\, Y}\ \mathrm{R}g^{+}$$

$$\dfrac{}{f^{+}\, X \underset{g}{\Longrightarrow} g^{+}\, Y}\ \mathrm{L}f^{+}$$

$$\dfrac{}{f^{-}\, f^{+}\, X \underset{fg}{\Longrightarrow} g^{+}\, Y}\ \mathrm{L}f^{-}$$

$$\dfrac{\dfrac{}{f^{-}\, f^{+}\, X \underset{h}{\Longrightarrow} g^{+}\, Y}}{f^{-}\, f^{+}\, X \underset{\mathrm{id}_A}{\Longrightarrow} h^{-}\, g^{+}\, Y}\ \mathrm{R}h^{-}$$



Formulas

Judgment $S \sqsubset A$ ("S an object of $\mathcal{B}\mathrm{if}(p)$ lying over A")

Formation rules:

$$\frac{X \in \mathcal{D} \quad p(X) = A}{X \sqsubset A} \qquad \frac{S \sqsubset A \quad f : A \to B}{f^{+}\, S \sqsubset B} \qquad \frac{f : A \to B \quad T \sqsubset B}{f^{-}\, T \sqsubset A}$$

Derivations

Judgment $\alpha : S \Longrightarrow_h T$ ("$\alpha : S \to T$ represents an arrow of $\mathcal{B}\mathrm{if}(p)$ lying over h")

Inference rules:

$$\frac{\delta : X \to Y \in \mathcal{D} \qquad p(\delta) = f}{\delta : X \underset{f}{\Longrightarrow} Y}\ \delta$$

$$\frac{\alpha : T \underset{g'}{\Longrightarrow} T'}{\overline{g}.\alpha : g^- T \underset{gg'}{\Longrightarrow} T'}\ \mathrm{L}g^- \qquad \frac{\alpha : S' \underset{f'}{\Longrightarrow} S}{\alpha.f : S' \underset{f'f}{\Longrightarrow} f^+ S}\ \mathrm{R}f^+$$

$$\frac{\alpha : S \underset{fg}{\Longrightarrow} T}{f \backslash_g \alpha : f^+ S \underset{g}{\Longrightarrow} T}\ \mathrm{L}f^+ \qquad \frac{\alpha : S \underset{fg}{\Longrightarrow} T}{\alpha\ {}_f/\overline{g} : S \underset{f}{\Longrightarrow} g^- T}\ \mathrm{R}g^-$$

Permutation equivalences (in proof-term syntax)

$$(\overline{f}.\alpha).h \sim \overline{f}.(\alpha.h) \qquad \text{for } \alpha \text{ over } g \tag{1}$$

$$(f\backslash_g\, \alpha).h \sim f\backslash_{gh}\, (\alpha.h) \qquad \text{for } \alpha \text{ over } fg \tag{2}$$

$$(\overline{f}.\alpha)\ {}_{fg}/\,\overline{h} \sim \overline{f}.(\alpha\ {}_g/\,\overline{h}) \qquad \text{for } \alpha \text{ over } gh \tag{3}$$

$$(f\backslash_{gh}\, \alpha)\ {}_g/\,\overline{h} \sim f\backslash_g\, (\alpha\ {}_{fg}/\,\overline{h}) \qquad \text{for } \alpha \text{ over } fgh \tag{4}$$

Identity and Composition

Identity derivations defined by induction on the formula:

$$\frac{S \sqsubset A}{S \underset{\mathrm{id}_A}{\Longrightarrow} S}\ \mathrm{id}_S \qquad \mathrm{id}_{f+\,S} \stackrel{\mathrm{def}}{=} f\backslash_{\mathrm{id}_B}\,(\mathrm{id}_S.f) \qquad \mathrm{id}_{g-\,T} \stackrel{\mathrm{def}}{=} (\overline{g}.\mathrm{id}_T)\,{}_{\mathrm{id}_B}/\,\overline{g}$$

Composition is cut-elimination:

$$\frac{\alpha : S \underset{g}{\Longrightarrow} T \qquad \beta : T \underset{h}{\Longrightarrow} U}{\alpha \cdot \beta : S \underset{gh}{\Longrightarrow} U}$$

Identity and Composition

Principal cuts:

$$
\begin{array}{rcl}
(\alpha.f) \cdot (f \backslash_h \beta) & \stackrel{\text{def}}{=} & \alpha \cdot \beta \\
(\alpha\ {}_g/\ \overline{f}) \cdot (\overline{f}.\beta) & \stackrel{\text{def}}{=} & \alpha \cdot \beta
\end{array}
$$

Commutative cuts:

$$
\begin{array}{rcl}
(\overline{f}.\alpha) \cdot \beta & \stackrel{\text{def}}{=} & \overline{f}.(\alpha \cdot \beta) \\
(f \backslash_g \alpha) \cdot \beta & \stackrel{\text{def}}{=} & f \backslash_{gh} (\alpha \cdot \beta)
\end{array}
\qquad
\begin{array}{rcl}
\alpha \cdot (\beta.f) & \stackrel{\text{def}}{=} & (\alpha \cdot \beta).f \\
\alpha \cdot \beta\ {}_h/\ \overline{f} & \stackrel{\text{def}}{=} & (\alpha \cdot \beta)\ {}_{gh}/\ \overline{f}
\end{array}
$$

Note: ambiguous cases up to equivalence.

Putting it all together

Let $\mathcal{B}\mathrm{if}(p)$ be the category whose objects are bifibrational formulas and whose arrows are $(\sim)$-equivalence classes of derivations.

Let Λ_p be the functor $\mathcal{B}\mathrm{if}(p) \to \mathcal{C}$ sending $(S \sqsubset A)$ to A and $(\alpha : S \Longrightarrow_f T)$ to f.

Theorem. $\Lambda_p : \mathcal{B}\mathrm{if}(p) \to \mathcal{C}$ is the free bifibration on $p : \mathcal{D} \to \mathcal{C}$.

Putting it all together

Let $\mathcal{B}\mathrm{if}(p)$ be the category whose objects are bifibrational formulas and whose arrows are $(\sim)$-equivalence classes of derivations.

Let Λ_p be the functor $\mathcal{B}\mathrm{if}(p) \to \mathcal{C}$ sending $(S \sqsubset A)$ to A and $(\alpha : S \Longrightarrow_f T)$ to f.

Theorem. $\Lambda_p : \mathcal{B}\mathrm{if}(p) \to \mathcal{C}$ is the free bifibration on $p : \mathcal{D} \to \mathcal{C}$.

Note the construction easily and usefully generalizes to $(\mathcal{P}, \mathcal{N})$-fibrations, where one restricts the pushforward and pullback operations to some classes of arrows:

$$\frac{S \sqsubset A \qquad f : A \to B \in \mathcal{P}}{f^+ S \sqsubset B} \qquad\qquad \frac{g : B \to C \in \mathcal{N} \qquad T \sqsubset C}{g^- T \sqsubset B}$$

Canonicity

In general, permutation equivalence is undecidable! But using ideas from proof theory, we can derive unique normal forms for $\mathcal{B}\mathrm{if}(p)$ when $\mathcal{C}$ is *factorization preordered* (FP), or more generally for $\mathcal{B}\mathrm{if}(p, \mathcal{P}, \mathcal{N})$ when $\mathcal{C}$ is both $\mathcal{P}$-FP and $\mathcal{N}$-FP.

A category is $\mathcal{W}$-**factorization preordered** just in case every commuting square of the following form has at most one diagonal filler:

$$\begin{tikzcd}
\cdot \arrow[r, "x"] \arrow[d, "f \in \mathcal{W}"'] & \cdot \arrow[d, "k \in \mathcal{W}"] \\
\cdot \arrow[r, "y"'] \arrow[ur, dashed, "g = h"] & \cdot
\end{tikzcd}$$

Canonicity gives us both decidability and enumeration!

5. Three examples of free bifibrations

Example #1

Let $p_2 : 1 \to 2$ be the functor shown below:

$$\begin{tikzcd} 1 \arrow[d, "p_2"'] & * \\ 2 & 0 \arrow[r, "f"] & 1 \end{tikzcd}$$

Question: what is the fiber of $\mathcal{B}\mathrm{if}(p_2)$ over 0?

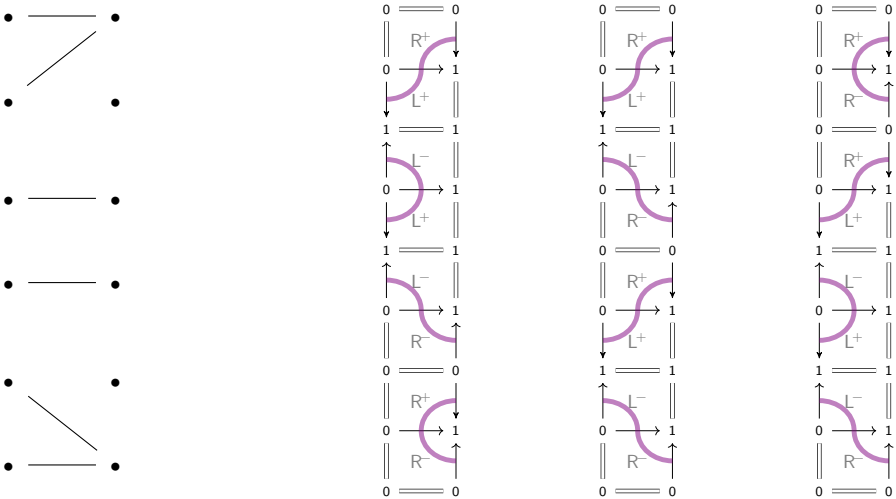
Example #1

Let $p_2 : 1 \to 2$ be the functor shown below:

$$\begin{tikzcd} 1 \arrow[d, "p_2"'] & * \\ 2 & 0 \arrow[r, "f"] & 1 \end{tikzcd}$$

Question: what is the fiber of $\mathcal{B}\mathrm{if}(p_2)$ over 0?

Answer: $\mathcal{B}\mathrm{if}(p_2)_0 \simeq \Delta$, the category of finite ordinals and order-preserving functions!


0 0
R+
0 1
L+
1 1
L−
0 1
L+
1 1
L−
0 1
R−
0 0
R+
0 1
R−
0 0
R+
0 1
L+
1 1
L−
0 1
R−
0 0
R+
0 1
L+
1 1
L−
0 1
R−
0 0
R+
0 1
R−
0 0
R+
0 1
L+
1 1
L−
0 1
L+
1 1
L−
0 1
R−
0 0

Example #2

Let ω be the totally ordered set of natural numbers considered as a category.

$$\begin{tikzcd} 1 \arrow[d, "p_\omega"'] & * \\ \omega & 0 \arrow[r, "f_0"] & 1 \arrow[r, "f_1"] & 2 \arrow[r, "f_2"] & \cdots \end{tikzcd}$$

Question: what is the fiber of $\mathcal{B}\mathrm{if}(p_\omega)$ over 0?

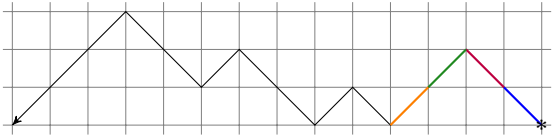
Example #2

Let ω be the totally ordered set of natural numbers considered as a category.

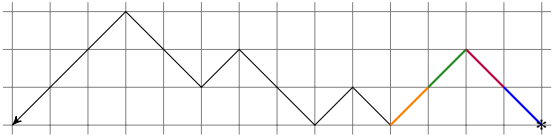
$$\begin{tikzcd} 1 \arrow[d, "p_\omega"'] & * \\ \omega & 0 \arrow[r, "f_0"] & 1 \arrow[r, "f_1"] & 2 \arrow[r, "f_2"] & \cdots \end{tikzcd}$$

Question: what is the fiber of $\mathcal{B}\mathrm{if}(p_\omega)$ over 0?

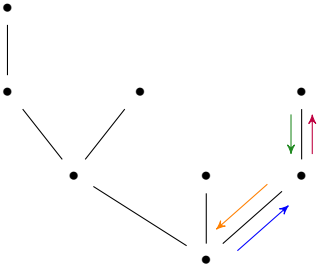
Answer: $\mathcal{B}\mathrm{if}(p_2)_0 \simeq \mathsf{PTree}$, the subcategory of $[\omega^{\mathrm{op}}, \Delta]$ spanned by finite plane trees!

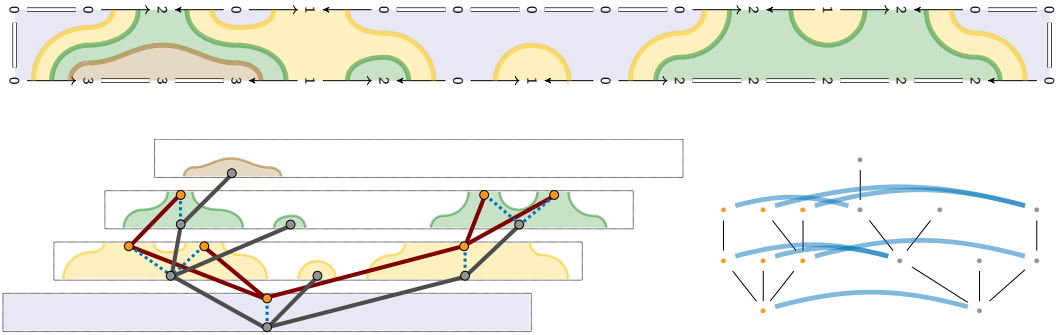


$f^- \; f^- \; f^- \; f^+ \; f^+ \; f^- \; f^+ \; f^+ \; f^- \; f^+ \; f^- \; f^- \; f^+ \; f^+ \; *$



$f^- \; f^- \; f^- \; f^+ \; f^+ \; f^- \; f^+ \; f^+ \; f^- \; f^+ \; f^- \; f^- \; f^+ \; f^+ \; *$





Example #3

Let $i : \mathbb{N} \to \Delta$ be the inclusion of the natural numbers (seen as a discrete category) into the simplex category, and build the free **(epi,mono)-fibration** generated by i.

Example #3

Let $i : \mathbb{N} \to \Delta$ be the inclusion of the natural numbers (seen as a discrete category) into the simplex category, and build the free **(epi,mono)-fibration** generated by i.

Next quotient by the <u>Beck-Chevalley condition for bicartesian squares</u>:

$$\begin{tikzcd} \cdot \arrow[r, two heads, "c"] \arrow[d, tail, "a"'] \arrow[dr, phantom, "\mathrm{BC}"] & \cdot \arrow[d, tail, "d"] \\ \cdot \arrow[r, two heads, "b"'] & \cdot \end{tikzcd} \qquad c^+ \, a^- \overset{\sim}{\Longrightarrow} d^- \, b^+$$

Example #3

Let $i : \mathbb{N} \to \Delta$ be the inclusion of the natural numbers (seen as a discrete category) into the simplex category, and build the free **(epi,mono)-fibration** generated by i.

Next quotient by the <u>Beck-Chevalley condition for bicartesian squares</u>:

$$\begin{tikzcd} \cdot \arrow[r, two heads, "c"] \arrow[d, tail, "a"'] \arrow[dr, phantom, "\mathrm{BC}"] & \cdot \arrow[d, tail, "d"] \\ \cdot \arrow[r, two heads, "b"'] & \cdot \end{tikzcd} \qquad c^+ \, a^- \overset{\sim}{\Longrightarrow} d^- \, b^+$$

Question: what does this give you?

Example #3

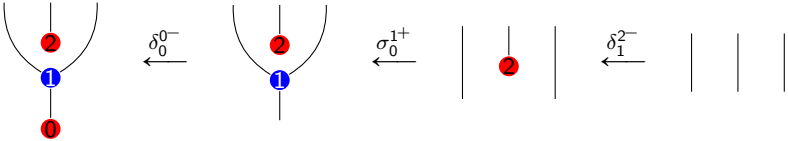
Let $i : \mathbb{N} \to \Delta$ be the inclusion of the natural numbers (seen as a discrete category) into the simplex category, and build the free **(epi,mono)-fibration** generated by i.

Next quotient by the <u>Beck-Chevalley condition for bicartesian squares</u>:

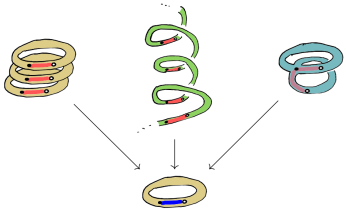
$$\begin{tikzcd} \cdot \arrow[r, two heads, "c"] \arrow[d, tail, "a"'] \arrow[dr, phantom, "\mathrm{BC}"] & \cdot \arrow[d, tail, "d"] \\ \cdot \arrow[r, two heads, "b"'] & \cdot \end{tikzcd} \qquad c^+ \, a^- \overset{\sim}{\Longrightarrow} d^- \, b^+$$

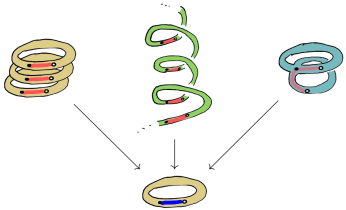
Question: what does this give you? Answer: the lattice of noncrossing partitions!

$\delta_0^{0-}\ \sigma_0^{1+}\ \delta_1^{2-}\ 3:$

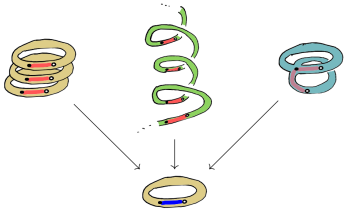


Conclusion



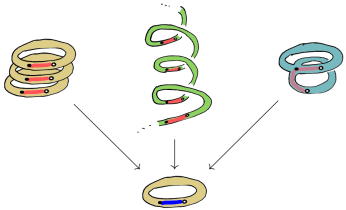


A variety of computational and deductive systems may be considered as bundles of categories, in a way that allows to treat them uniformly.



A variety of computational and deductive systems may be considered as bundles of categories, in a way that allows to treat them uniformly.

Fibrational properties as well as free constructions play an important role, and have deep connections to logic.



A variety of computational and deductive systems may be considered as bundles of categories, in a way that allows to treat them uniformly.

Fibrational properties as well as free constructions play an important role, and have deep connections to logic.

References:

- w/P.-A. Melliès: "Functors are type refinement systems" (POPL 2015); "The categorical contours of the Ch.-Sch. repr. theorem" (LMCS 21:2, 2025)
- w/B. Clarke & G. Scherer: "The free bifibration on a functor", arxiv:2511.07314